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FINAL REPORT

**Analysis of Airflow in the Volvo Bluff-Body Experiment : Turbulence
Model Validation and Proper Orthogonal Decomposition**

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Analysis of Airflow in the Volvo Bluff-Body Experiment : Turbulence Model Validation and Proper Orthogonal Decomposition

Abstract:

This study investigates the airflow behavior around a bluff-body geometry inspired by the Volvo bluff-body burner experiment. The focus is on the non-reacting case with a bulk inlet velocity of 16.6 m/s. Two turbulence models — $k-\epsilon$ and $k-\omega$ SST — were implemented in OpenFOAM to simulate the flow field and compare their performance against experimental data. To improve spatial resolution, Adaptive Mesh Refinement (AMR) based on vorticity magnitude was applied. Additionally, Proper Orthogonal Decomposition (POD) was used to extract dominant coherent structures from the velocity field. The first few POD modes captured the majority of the flow energy, indicating low-dimensional behavior of the dominant dynamics. This analysis aids in understanding turbulence structures and supports the development of reduced-order models for complex flows.

1. Introduction

1.1. Background and Motivation

Bluff-body flows are widely encountered in combustion systems due to their ability to create recirculation zones that help stabilize flames. Accurate simulation of such flows is essential for improving combustion efficiency and reducing emissions.

The Volvo bluff-body experiment provides a reliable benchmark for validating numerical models, as it offers a simple geometry with detailed experimental data. In this study, the focus is on the non-reacting case to isolate the aerodynamic behavior of the flow.

Common turbulence models like $k-\epsilon$ and $k-\omega$ SST are evaluated in OpenFOAM for their predictive accuracy. To improve resolution in critical flow regions, Adaptive Mesh Refinement (AMR) is used based on vorticity magnitude. Additionally, Proper Orthogonal Decomposition (POD) helps identify dominant flow structures and supports reduced-order modeling.

1.2. Objectives of the Study

To numerically simulate airflow around a bluff-body (specifically, a prism) based on the Volvo bluff-body burner experiment. The focus is on the non-reacting flow case with an inlet velocity of 16.6 m/s. The primary objectives are:

1. To evaluate which turbulence model implemented in OpenFOAM best replicates the experimental data.
2. To explore Adaptive Mesh Refinement (AMR) for improved spatial resolution in flow regions of interest.
3. To apply Proper Orthogonal Decomposition (POD) to extract dominant flow structures based on their energy content.

1.3. Overview of Volvo Bluff-Body Experiment

The Volvo burner features the canonical configuration of a bluff-body stabilized premixed flame, which is fundamental setup to study flame anchoring and instabilities under lean combustion conditons. Lean combustion is desirable for high efficiency and low emissions, but it suffers from flame instability and lean blowout (LBO) risks .This configuration was studied experimentally under the Volvo Flygmotor AB program, through which extensive measurements wre generated using techniques like LDA and CARS. Only non-reacting case has been considered in this study.

2.Case Description

2.1. Computational Domain

The model was developed using SolidWorks. The computational domain employed in this study is illustrated in *Figure 2*. The domain has a depth equal to twice the width of the bluff body and corresponds to one-third of the experimental channel length. The complete combustor section located downstream of the flameholder is included in the domain. A triangular bluff body with each side measuring 0.04 m is positioned 0.2 m downstream from the inlet.

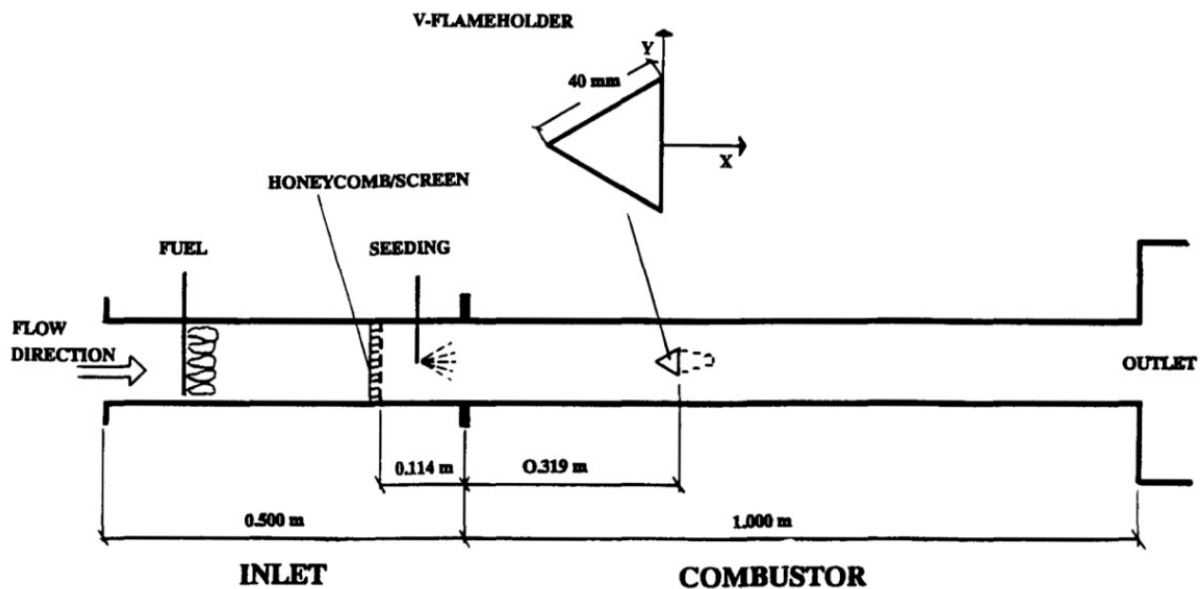


Figure 1: Figure 1: Schematic of the Volvo test rig (Sjunneson et al. [1])

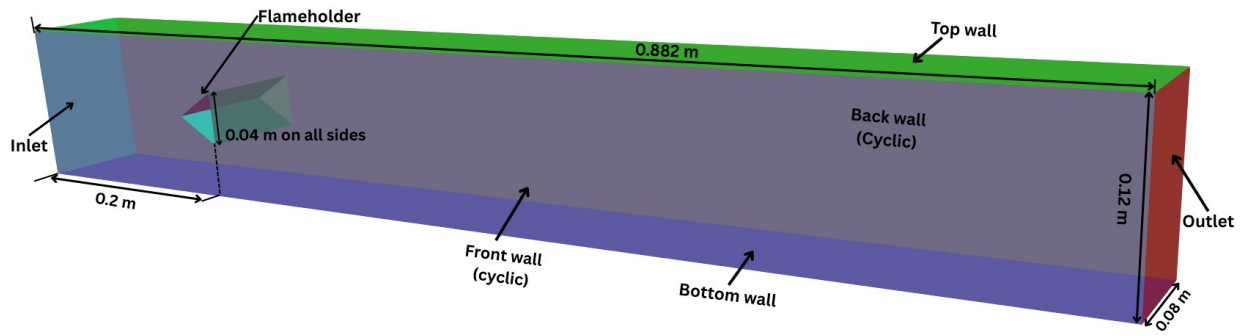


Figure 2: Figure 2 : Computational domain and boundary Conditions

2.2. Meshing Strategy

The fluid domain around the bluff body was discretized using a structured hexahedral mesh generated in **ANSYS Meshing**. The mesh comprises a total of **535,928 hexahedral cells**, with no inclusion of other cell types such as tetrahedra or prisms. The total number of nodes in the mesh is **565,536**, of which **507,090** are located in the interior of the domain.

To ensure adequate resolution in regions of high flow gradients, mesh biasing was applied near the bluff body and walls. This provided finer cells in the near-wall and wake regions, while maintaining coarser cells in the far-field to optimize computational cost.

The mesh was exported to OpenFOAM format and validated using the checkMesh utility. All topological and geometrical checks passed successfully, confirming that the mesh is valid and suitable for accurate CFD simulations.

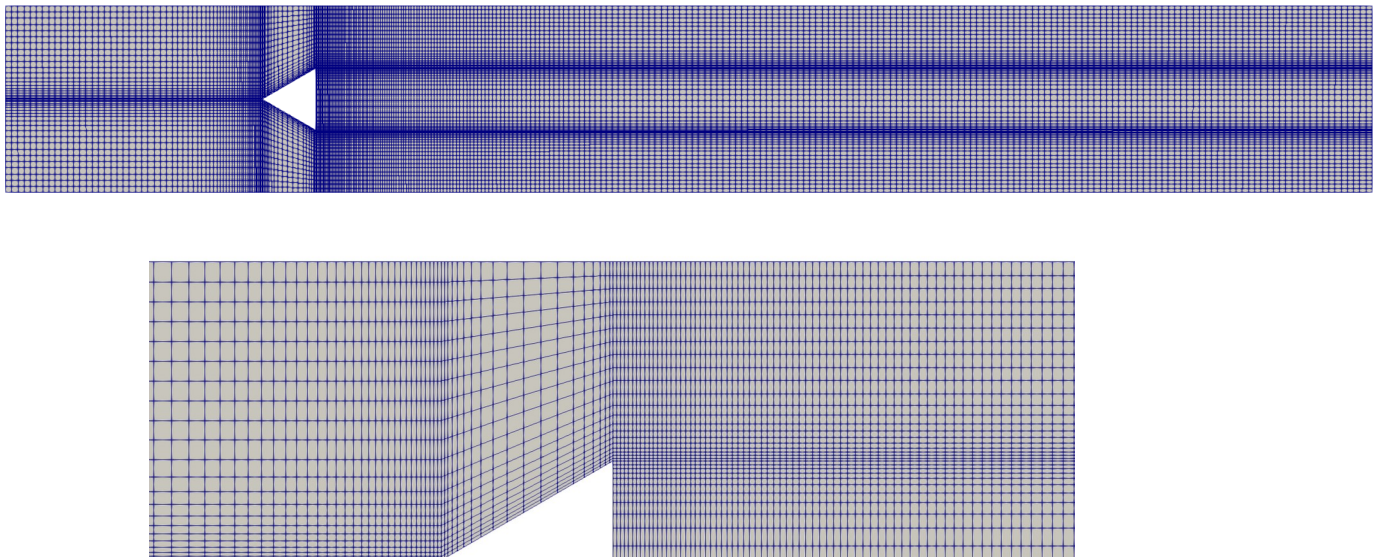


Figure 3: Structured Mesh

2.3. Boundary Conditions :

Cyclic boundary conditions are imposed in the spanwise direction on the front and back walls. A

uniform inlet velocity(U_{∞}) of 16.6 m/s is prescribed at the inlet, with a constant temperature of 288 K. The working fluid is air, and the flow is considered non-reacting. At the outlet, a pressure of 1 bar is specified using characteristic boundary conditions. No-slip, adiabatic wall conditions are applied to the flameholder as well as the top and bottom walls, in accordance with the guidelines provided in the MVP study [2].

Re	U_{bulk} [m/s]	Equivalence Ratio (Φ)	T_{in} [K]
48,000	16.6	0	288

Table 1: Operating Conditions

3. Numerical Setup:

3.1. Governing Equations:

The governing equations for incompressible flow under the RANS (Reynolds-Averaged Navier-Stokes) framework are:

Continuity Equation:

$$\nabla \cdot \vec{u} = 0$$

Momentum Equation:

$$\partial \vec{u} / \partial t + \vec{u} \cdot \nabla \vec{u} = -(1/\rho) \nabla p + \nu \nabla^2 \vec{u} - \nabla \cdot u_i' u_j'$$

Here, $u_i' u_j'$ represents the Reynolds stresses, which are modeled using turbulence models.

3.1.1. k-ε Model

The k-ε turbulence model is a two-equation model in which two transport equations are solved: one for the turbulent kinetic energy (k) and one for the turbulent dissipation rate (ε).

Transport Equations:

Turbulent Kinetic Energy (k):

$$\partial k / \partial t + u_j \partial k / \partial x_j = P_k - \epsilon + \partial / \partial x_j [(v + v_t / \sigma_k) \partial k / \partial x_j]$$

Turbulent Dissipation Rate (ε):

$$\partial \epsilon / \partial t + u_j \partial \epsilon / \partial x_j = C_{\epsilon_1} (\epsilon / k) P_k - C_{\epsilon_2} (\epsilon^2 / k) + \partial / \partial x_j [(v + v_t / \sigma_{\epsilon}) \partial \epsilon / \partial x_j]$$

Where:

- P_k is the production of turbulence
- $v_t = C_{\mu} (k^2 / \epsilon)$ is the eddy viscosity

Estimation of Initial Turbulence Quantities:

Assuming:

- $U = 16.6$ m/s (mean velocity)
- $l = 0.04$ m (turbulence length scale)

Turbulent kinetic energy:

$$k = (0.16 \times U)^2 = (0.16 \times 16.6)^2 = 1.03335 \text{ m}^2/\text{s}^2$$

Turbulent dissipation rate:

$$\epsilon = C_{\mu}^{3/4} \times k^{3/2} / l = 0.09^{3/4} \times (1.03335)^{3/2} / 0.04 = 61.644 \text{ m}^2/\text{s}^3$$

3.1.2. k-ω SST Model

The k-ω SST model solves for k (turbulent kinetic energy) and ω (specific dissipation rate). It blends the k-ε model in the free stream with the k-ω model near walls.

Transport Equations:

Turbulent Kinetic Energy (k):

$$\partial k / \partial t + u_j \partial k / \partial x_j = P_k - \beta^* k \omega + \partial / \partial x_j [(v + \sigma_k v_t) \partial k / \partial x_j]$$

Specific Dissipation Rate (ω):

$$\partial \omega / \partial t + u_j \partial \omega / \partial x_j = \alpha (\omega / k) P_k - \beta \omega^2 + \partial / \partial x_j [(v + \sigma_\omega v_t) \partial \omega / \partial x_j] + 2(1 - F_1) (\sigma_{\omega_2} / \omega) \partial k / \partial x_j \partial \omega / \partial x_j$$

Initial Estimate of ω :

$$\omega = \varepsilon / (C_\mu \times k) = 61.644 / (0.09 \times 1.03335) = 59.654$$

4. Results and Discussion

4.1. Turbulence Model Comparison:

4.1.1. k- ω SST Model Results:

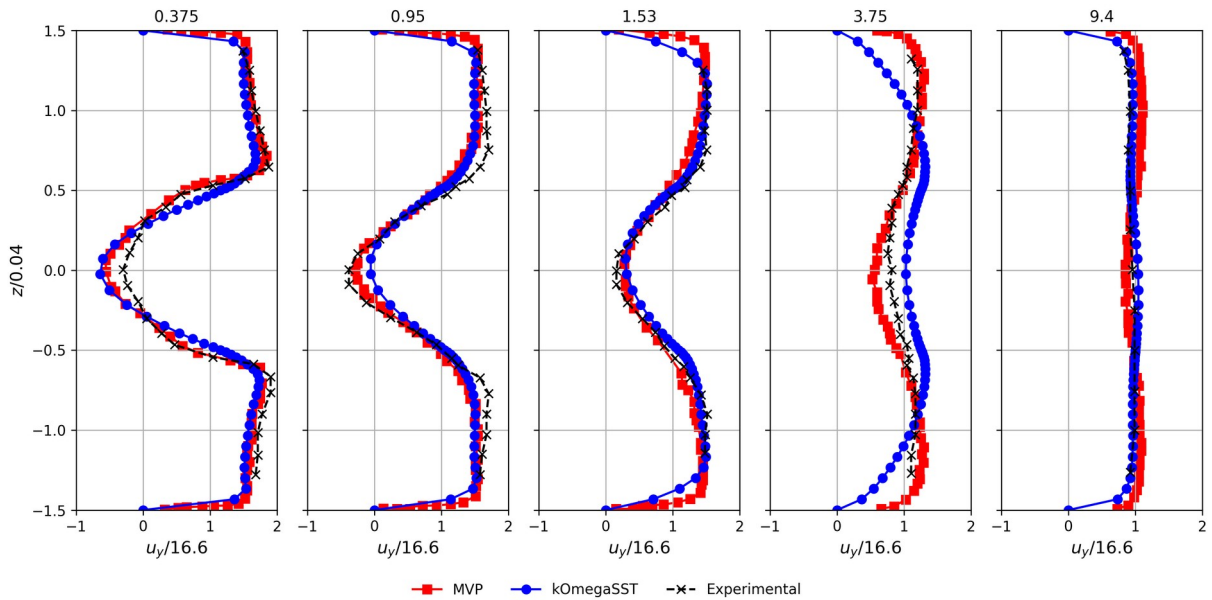


Figure 4: Normalized mean (dot) axial velocity profiles in comparison with experimental data (cross) and MVP study (square), at several axial locations ($y/D = 0.375, 0.95, 1.53, 3.75$ and 9.4 respectively)

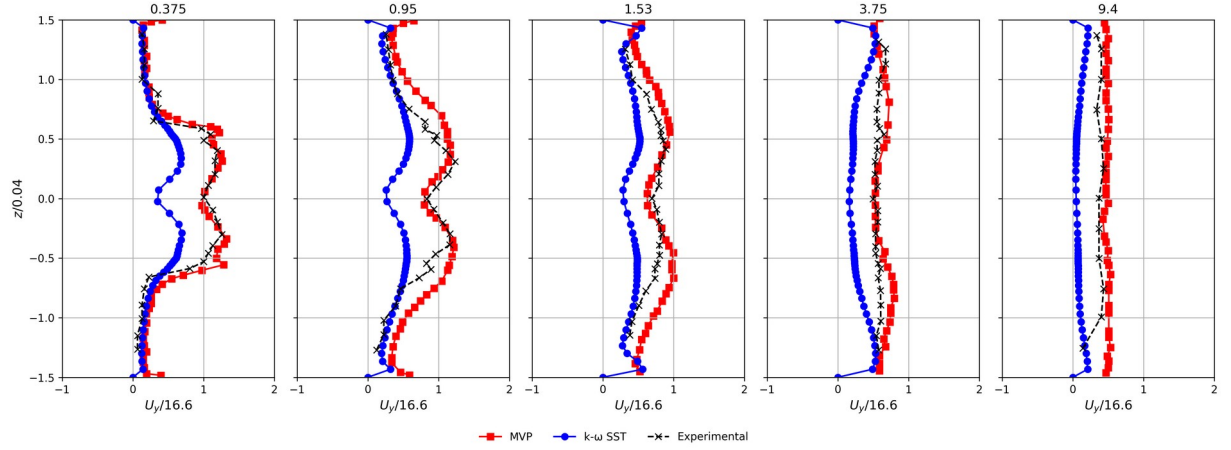


Figure 5: RMS (dot) axial velocity profiles in comparison with experimental data (cross) and MVP study (square), at several axial locations ($y/D = 0.375, 0.95, 1.53, 3.75$ and 9.4 respectively)

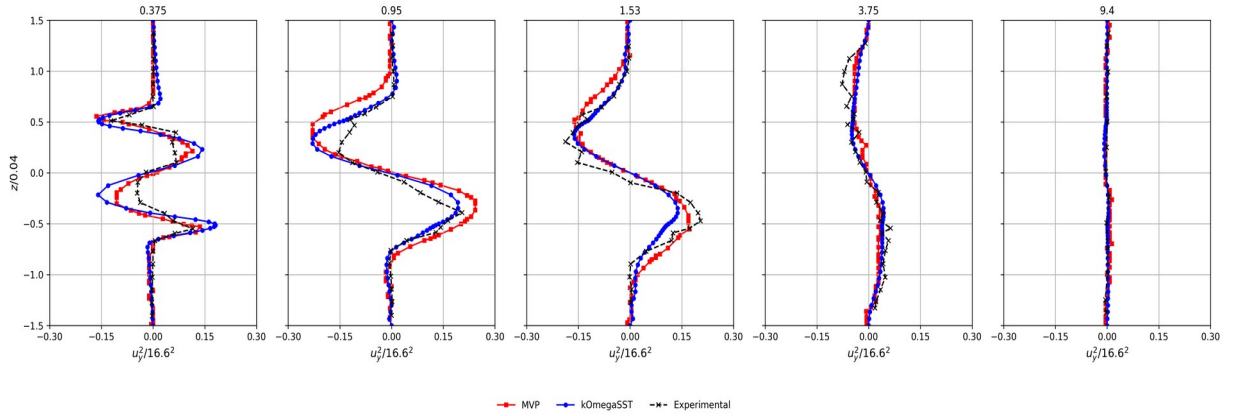


Figure 6: Normalized mean Reynolds stress profiles in comparison with experimental data (cross) and MVP study (square), at several axial locations ($y/D = 0.375, 0.95, 1.53, 3.75$ and 9.4 respectively)

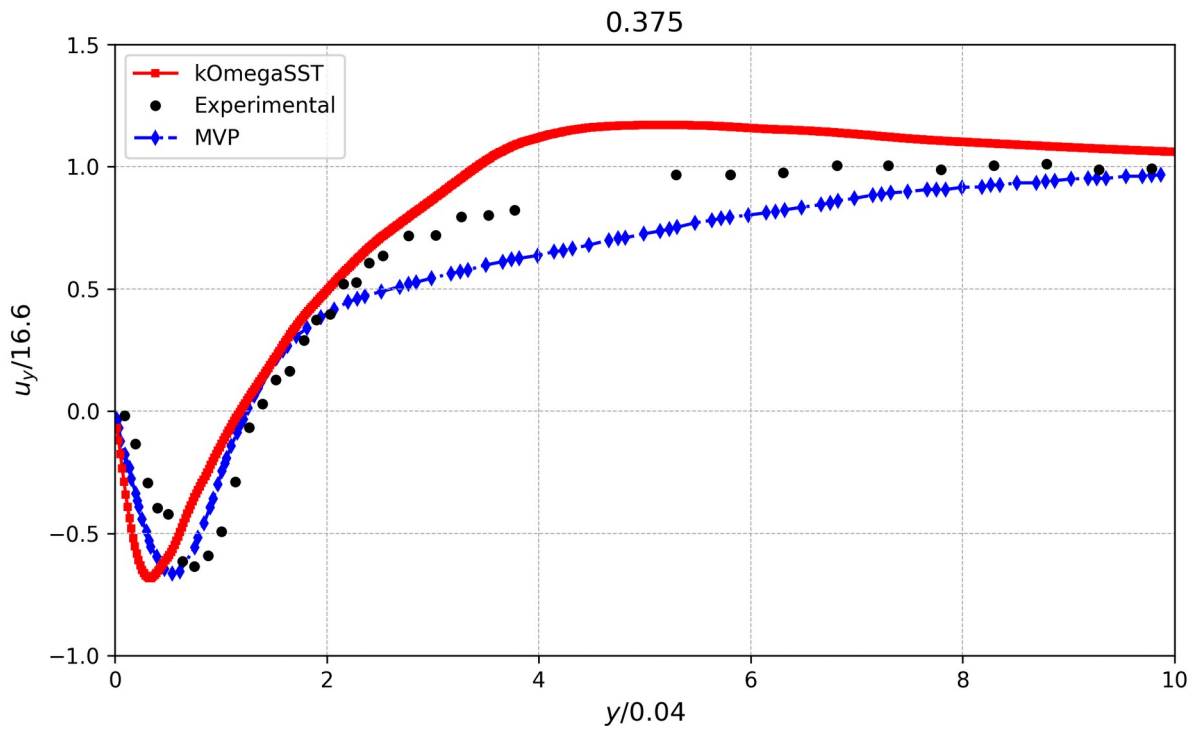


Figure 7: Centerline profiles for mean axial velocity

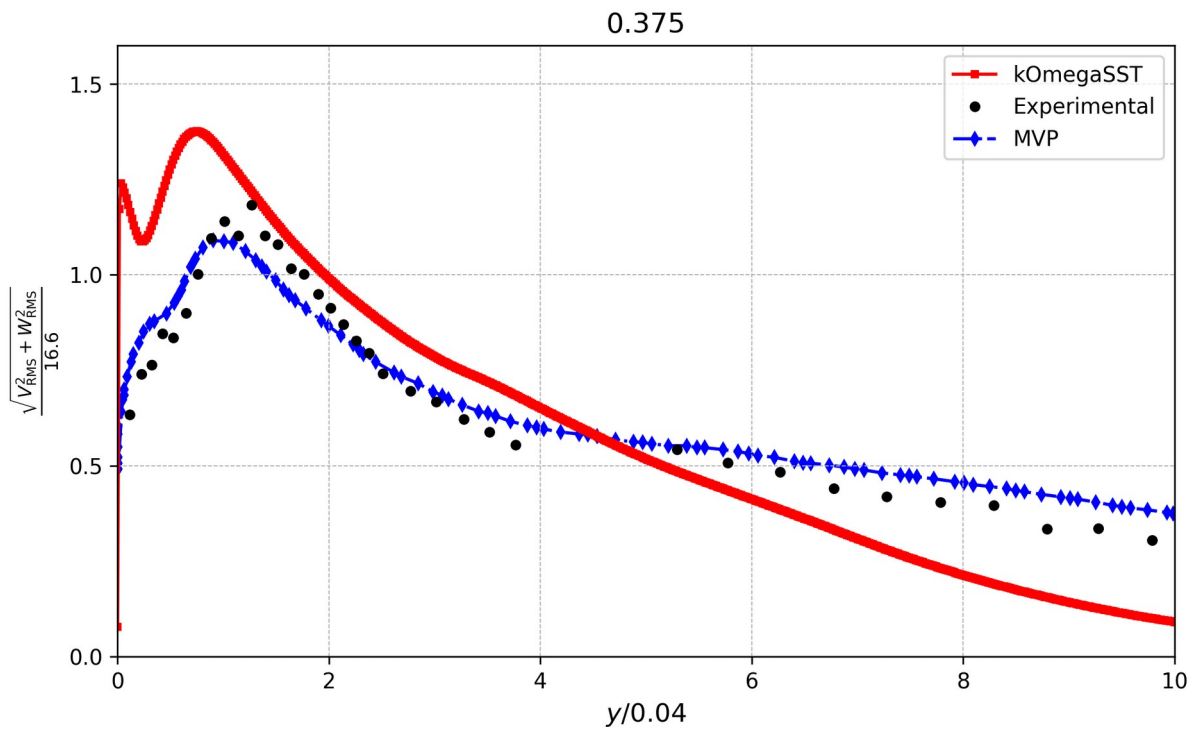


Fig.

4.1.2. k- ϵ Model Results:

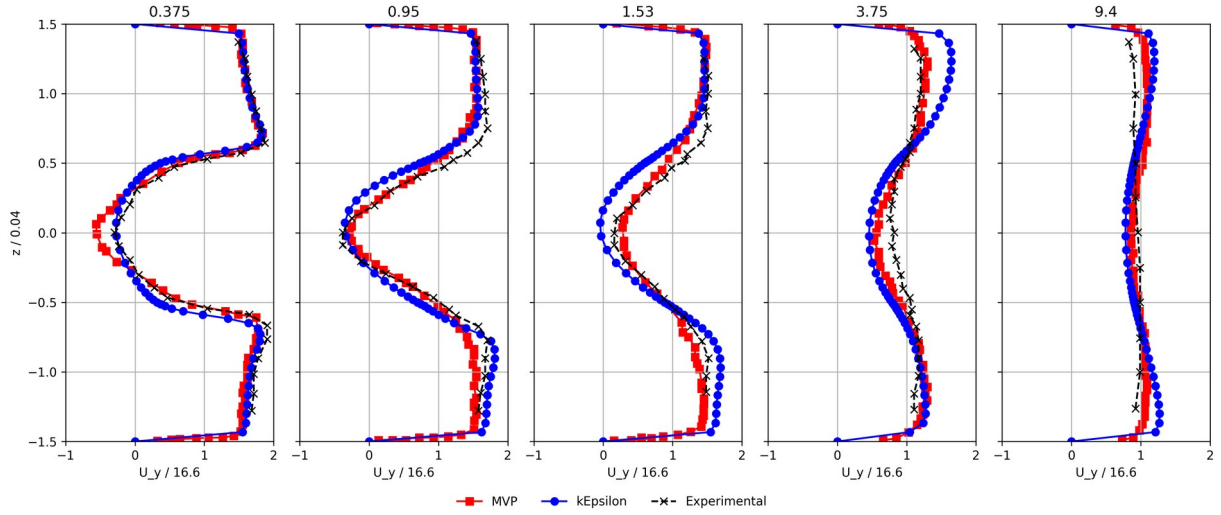


Figure 9: Normalized mean (dot) axial velocity profiles in comparison with experimental data (cross) and MVP study (square), at several axial locations ($y/D = 0.375, 0.95, 1.53, 3.75$ and 9.4 respectively)

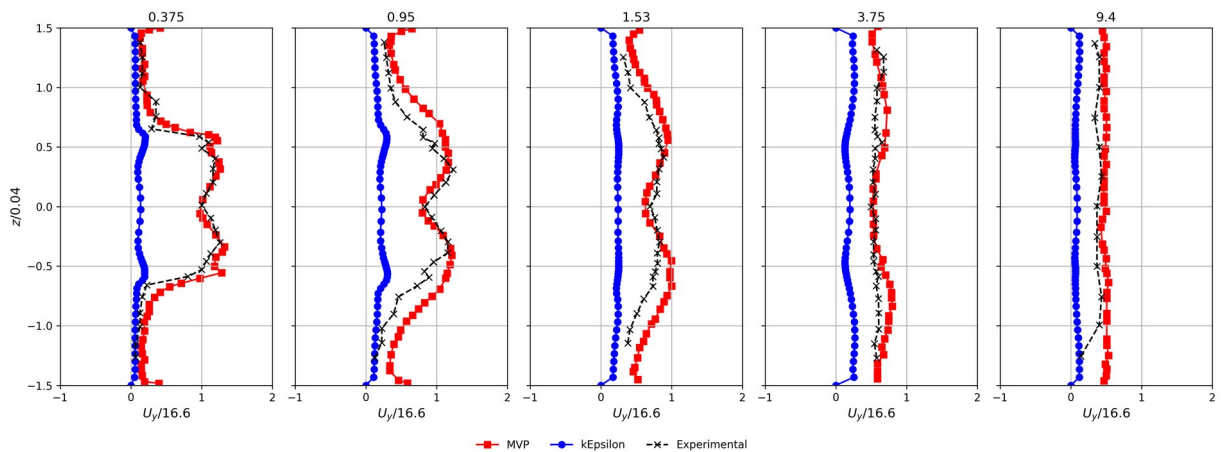


Figure 10: RMS (dot) axial velocity profiles in comparison with experimental data (cross) and MVP study (square), at several axial locations ($y/D = 0.375, 0.95, 1.53, 3.75$ and 9.4 respectively)

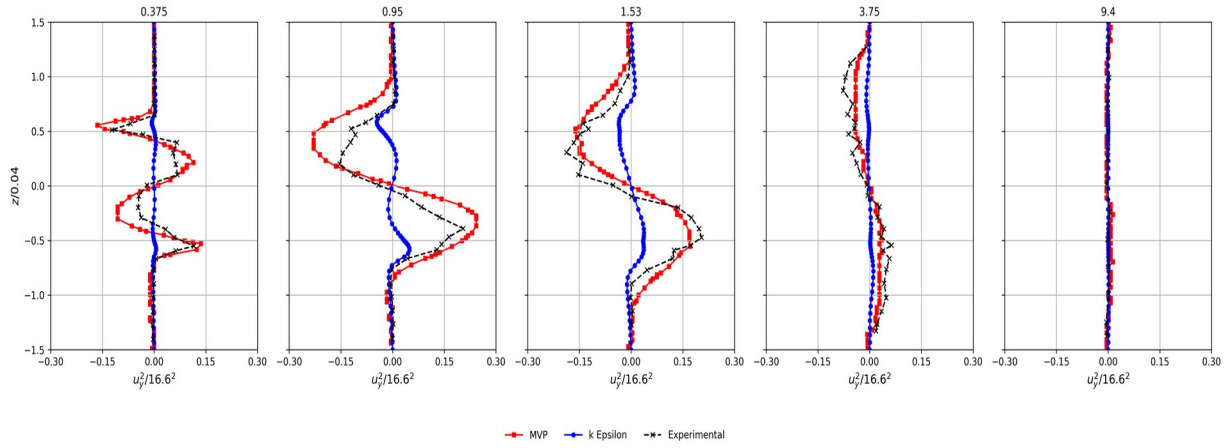


Figure 11: Normalized mean Reynolds stress profiles in comparison with experimental data (cross) and MVP study (square), at several axial locations($y/D = 0.375, 0.95, 1.53, 3.75$ and 9.4 respectively)

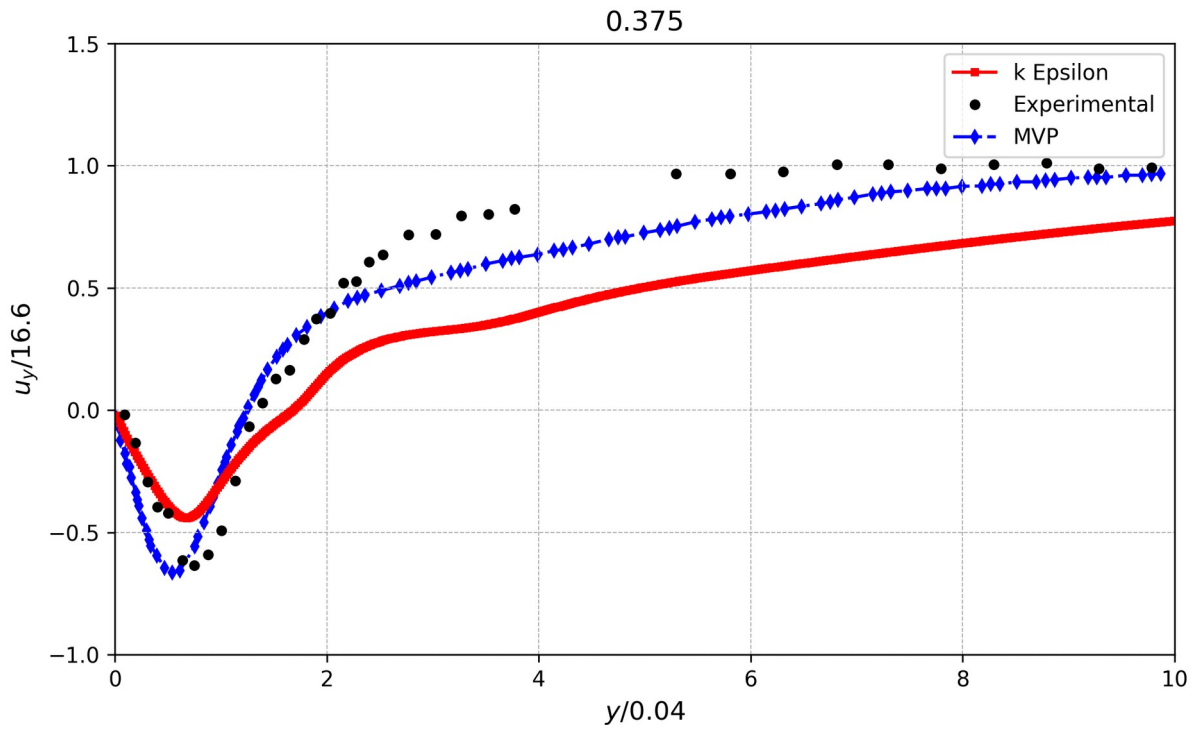


Figure 12: Centerline profiles for mean axial velocity

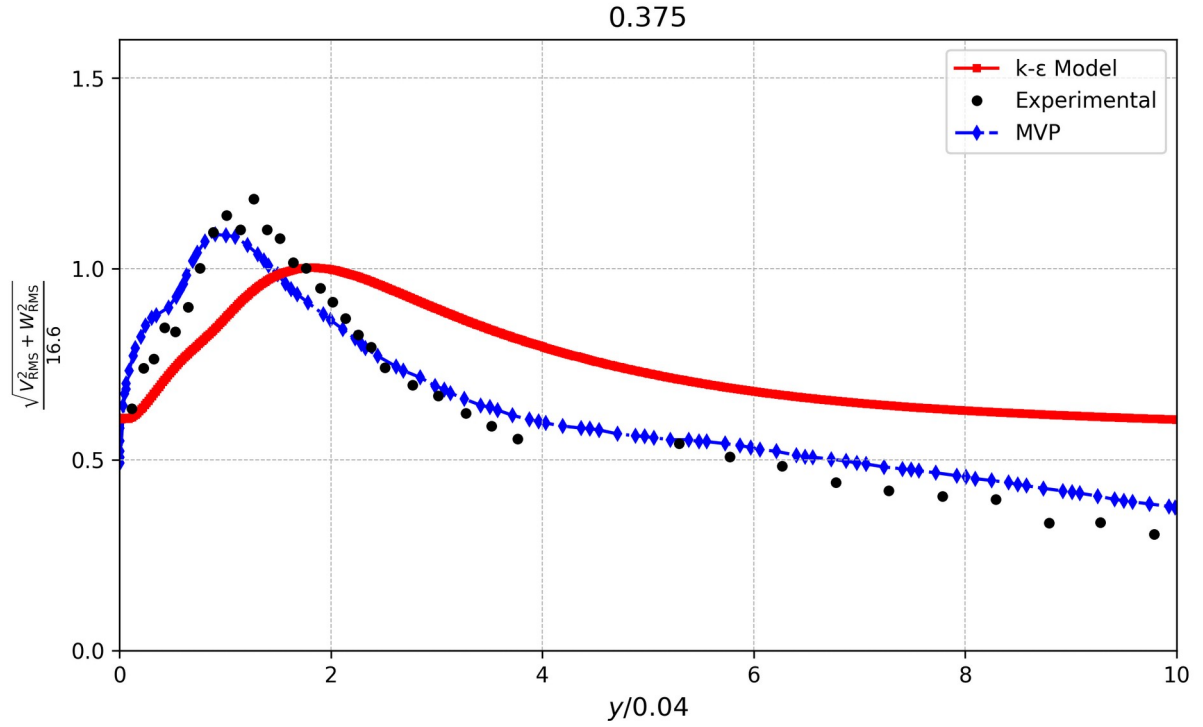


Figure 13: Centerline profiles for (a) mean axial velocity, (b) anisotropy, and (c) fluctuation

4.2. Turbulence Model Comparison (Errors):

Case	MAE (k- ω)	RMSE (k- ω)	MAE (k- ϵ)	RMSE (k- ϵ)	Better
0.375	0.206	0.234	0.161	0.236	k- ω
0.950	0.162	0.189	0.195	0.234	k- ω
1.530	0.094	0.111	0.221	0.273	k- ω
3.750	0.244	0.269	0.230	0.264	k- ω
9.400	0.037	0.043	0.178	0.205	k- ω
Average	0.149	0.169	0.197	0.242	k- ω

Table 2. Mean Axial Velocity

Case	MAE (k- ω)	RMSE (k- ω)	MAE (k- ϵ)	RMSE (k- ϵ)	Better
0.375	0.308	0.388	0.553	0.700	k- ω
0.950	0.310	0.402	0.515	0.596	k- ω
1.530	0.267	0.301	0.451	0.477	k- ω
3.750	0.286	0.303	0.380	0.383	k- ω
9.400	0.267	0.284	0.280	0.293	k- ω
Average	0.288	0.336	0.436	0.491	k- ω

Table 3. RMS Velocity

Case	MAE (k- ω)	RMSE (k- ω)	MAE (k- ϵ)	RMSE (k- ϵ)	Better
0.375	0.036	0.050	0.032	0.048	k- ω
0.950	0.023	0.035	0.057	0.082	k- ω

1.530	0.024	0.035	0.075	0.094	k- ω
3.750	0.012	0.015	0.037	0.040	k- ω
9.400	0.003	0.003	0.002	0.002	k- ω
Average	0.020	0.028	0.041	0.053	k- ω

Table 4. Reynolds Stresses

Model	MAE	RMSE
k- ϵ	0.237	0.274
k- ω SST	0.2	0.22

Table 4. U_y vs y

Model	MAE	RMSE
k- ϵ	0.237	0.274
k- ω SST	0.2	0.22

Table 5. Fluctuations

5. Proper Orthogonal Decomposition:

5.1. Introduction to POD

Proper Orthogonal Decomposition (POD) is a data-driven technique used to extract the dominant patterns or modes from complex datasets, especially in fluid dynamics, structural dynamics, and data compression.

Proper Orthogonal Decomposition is a mathematical method that:

- Breaks down a complex field (like velocity, pressure, etc.) into simpler, orthogonal modes.
- These modes are ranked based on their energy content or importance.
- It's similar to PCA (Principal Component Analysis) but applied to physical fields.

Dominant Flow Structures (Modes):

- POD extracts spatial patterns (modes) that represent coherent structures in the flow (like vortices or shear layers).

Energy Content of Each Mode:

- It tells how much "energy" (variance) is captured by each mode.
- Often, a few modes capture most of the dynamics, allowing for model simplification.

Time Coefficients:

- Shows how each mode evolves in time.

Reduced-Order Models (ROMs):

- POD enables the creation of simplified models that still capture the essential dynamics, useful for control or faster simulations.

5.2. POD Methodology:

Let's denote:

From time $t=20$ to $t=26.375$ sec with $\Delta t=0.025$ sec

- N = total number of time snapshot = 256 $\bar{u}$ u' [
- $t_1, t_2, \dots, t_N$ = discrete time steps (20, 20.025, 20.05, , 26.35, 26.375)

Let each velocity field at time t_i is a vector: $u(x, y, z, t_i) \in \mathbb{R}^M$,

Where:

- M = columns $\times$ rows of velocity matrix = total number of spatial points $\times$ 3 components (u, v, w)

The mean velocity field is:

$$\bar{u} = (1/N) \sum_i u(x, y, z, t_i)$$

Then the fluctuating field is:

$$u'(x, y, z, t_i) = u(x, y, z, t_i) - \bar{u}$$

The matrix B is constructed as:

$$B = [u'(t_1), u'(t_2), \dots, u'(t_N)] \in \mathbb{R}^{M \times N}$$

The correlation matrix is:

$$C = (1/N) B^T B \in \mathbb{R}^{N \times N}$$

Perform eigen-decomposition:

$C U = U \Lambda$, where Λ is a diagonal matrix of eigenvalues.

Spatial POD modes are computed by:

$$\Phi = B U \in \mathbb{R}^{M \times N}$$

Each mode ϕ_i is normalized as:

$$\hat{\phi}_i = \phi_i / \|\phi_i\|$$

Mode energy is computed as:

$$E_i = \lambda_i^2 / \sum_j \lambda_j^2$$

The time coefficients (amplitudes) for each mode are:

$$a_i(t_j) = \hat{\phi}_i^T \cdot u'(t_j)$$

Finally, the fluctuating field can be reconstructed as:

$$u'(x, y, z, t) \approx \sum_{i=1}^k a_i(t) \cdot \phi_i(x, y, z)$$

5.3. Vortex Shedding Analysis using FFT:

To identify the vortex shedding characteristics of the flow, Fast Fourier Transform (FFT) analysis was performed on the time history of the velocity signal downstream of the bluff body. The dominant frequency corresponding to vortex shedding was extracted using the Welch method.

The **dominant frequency** was found to be **1.7647 Hz**, corresponding to a **vortex shedding period** of **0.5667 s**. The **Strouhal number**, which is a dimensionless quantity characterizing vortex shedding, was computed using the formula:

$$St = U_b / (f \cdot d)$$

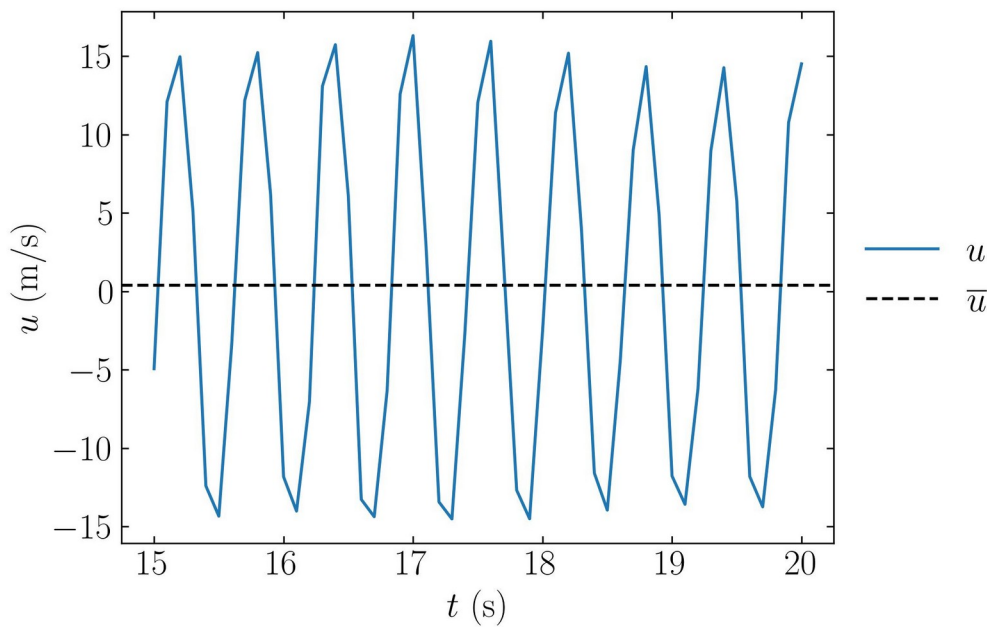
Where:

- $f = 1.7647$ Hz (dominant frequency)
- $d = 0.04$ m (characteristic length, flameholder side)
- $U_b = 16.6$ m/s (bulk velocity)

Substituting the values:

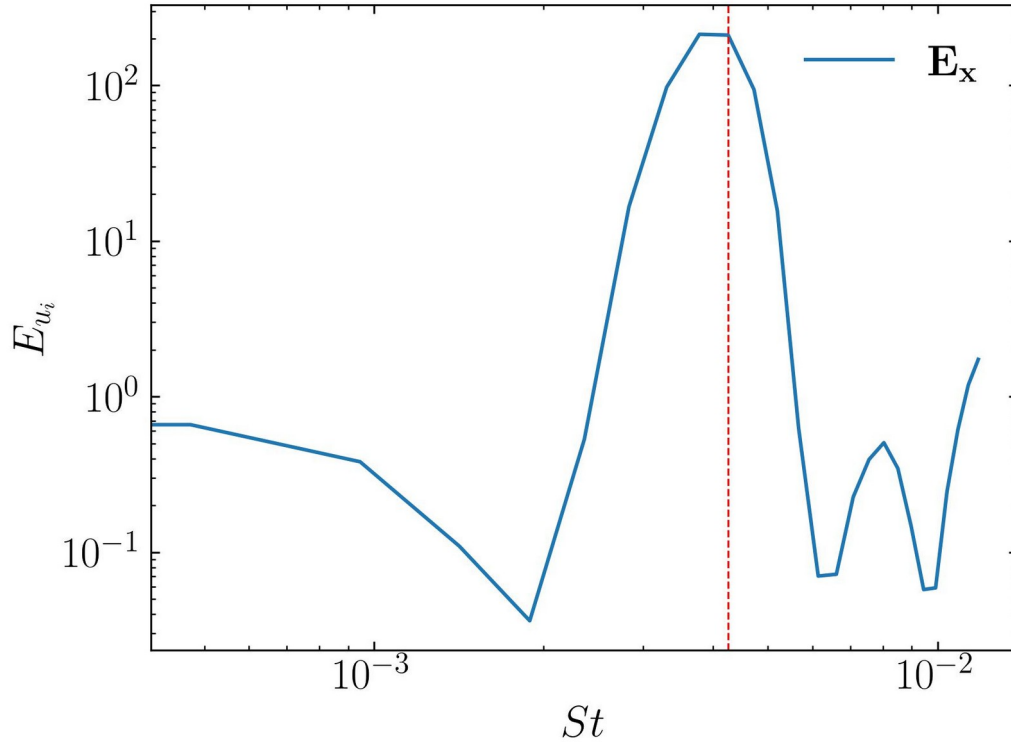
$$St = 16.6 / (1.7647 \times 0.04) \approx 0.004252$$

This low Strouhal number suggests a weak or low-frequency vortex shedding behavior, consistent with the bluff-body wake characteristics under the given flow conditions.



of u w.r.t. time

Figure 14: Variation



Figure

15: Power Spectral Density vs Strouhal Number

5.4 POD Results and Interpretation

The analysis was performed using 256 time-resolved velocity snapshots captured from $t = 20$ s to $t = 26.4$ s, with a uniform time interval of $\Delta t = 0.025$ s. This duration captures several vortex shedding cycles under the assumption of statistically steady and periodic unsteady flow.

For qualitative and quantitative interpretation, only the first 10 most energetic POD modes of the velocity vector were analyzed. The visualizations presented in this section correspond to a single representative time instant, $t = 26.375$ s, which lies within the periodic regime. These modes were chosen based on their significant energy contribution to the total flow dynamics.

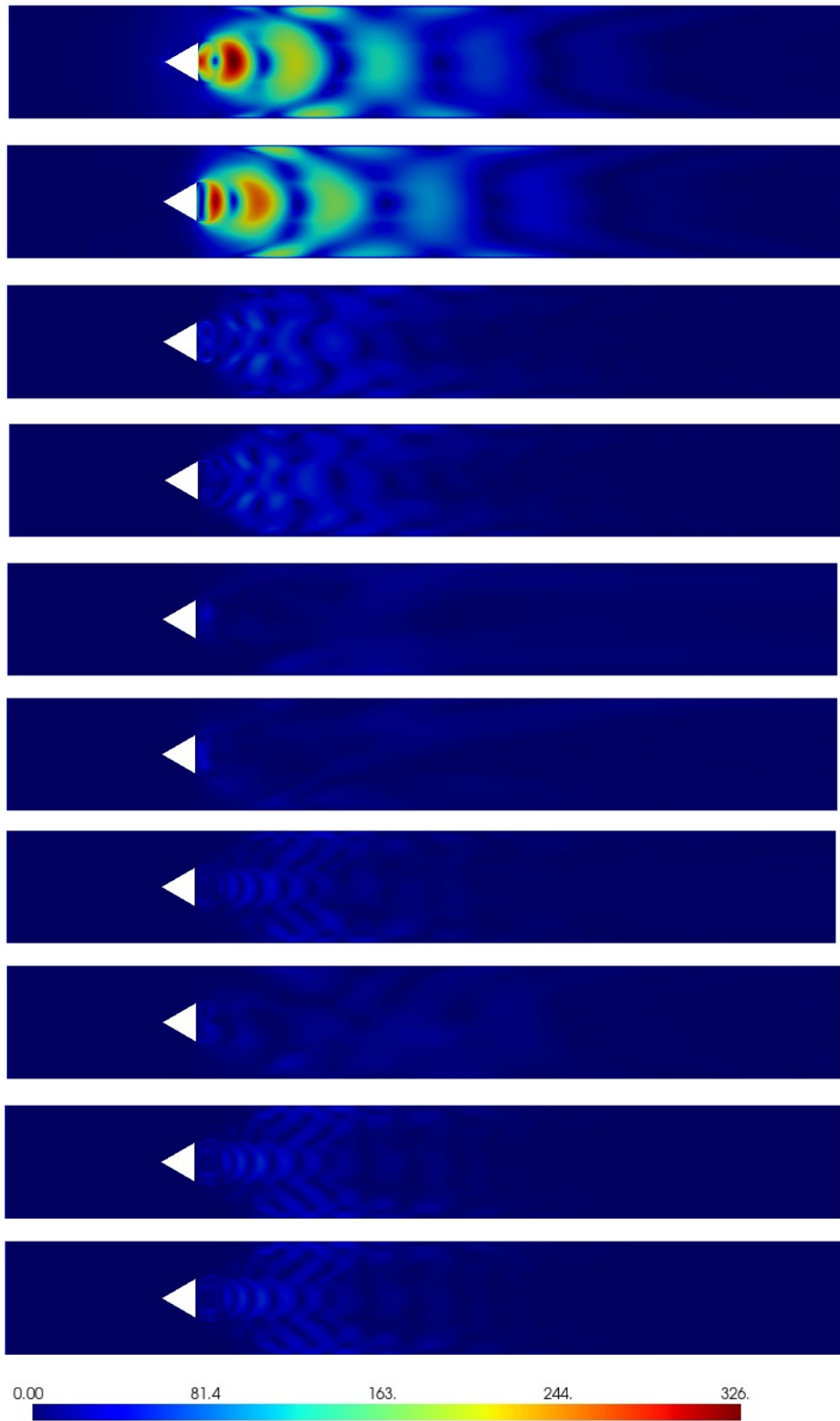
Mode Visualization

The spatial structures of the first 10 modes illustrate the dominant coherent flow features in the wake region. Each mode captures distinct spatial patterns associated with the alternating vortex shedding behind the bluff body. As the mode number increases, the spatial scales become finer, indicating the presence of smaller-scale turbulent structures.

Temporal Coefficients

To gain further insight into the temporal evolution of each mode, the amplitude (POD coefficient) vs. time plots were examined. These plots reveal oscillatory patterns for the leading modes, consistent with periodic vortex shedding behavior. The amplitude fluctuations diminish for higher-order modes, as they carry less energy.

5.4.1. Mode Shapes (Decomposed Modes):



Figure_16.Decomposed_Modes (Mode 1, 2, 3, 4, 5, 6, 7, 8, 9 & 10 respectively)

5.4.2.Flow Reconstruction and Validation:

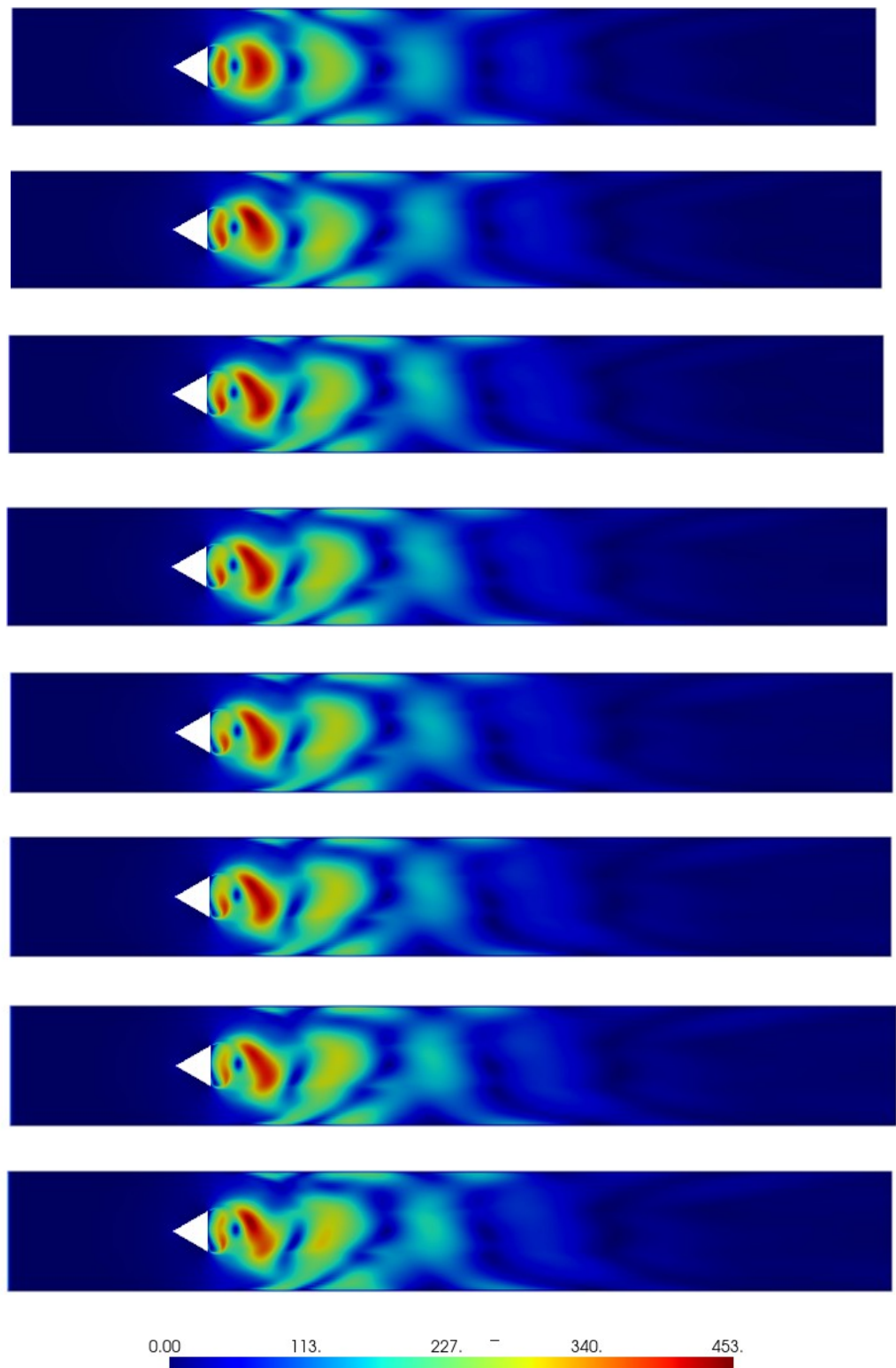
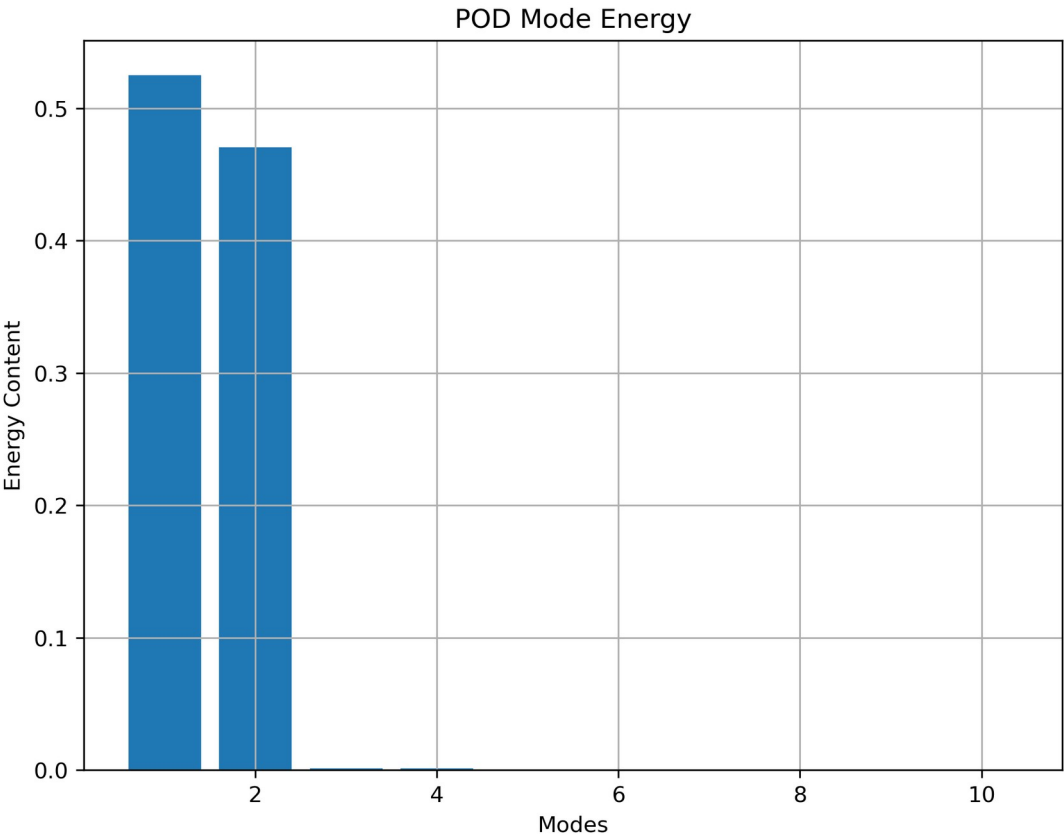


Figure 17. Reconstructed Velocity Contour(form first 2, 3, 4, 5, 6, 7, 8, 9 & 10 modes respectively)

5.4.3. Energy Content and Cumulative Energy of Modes :

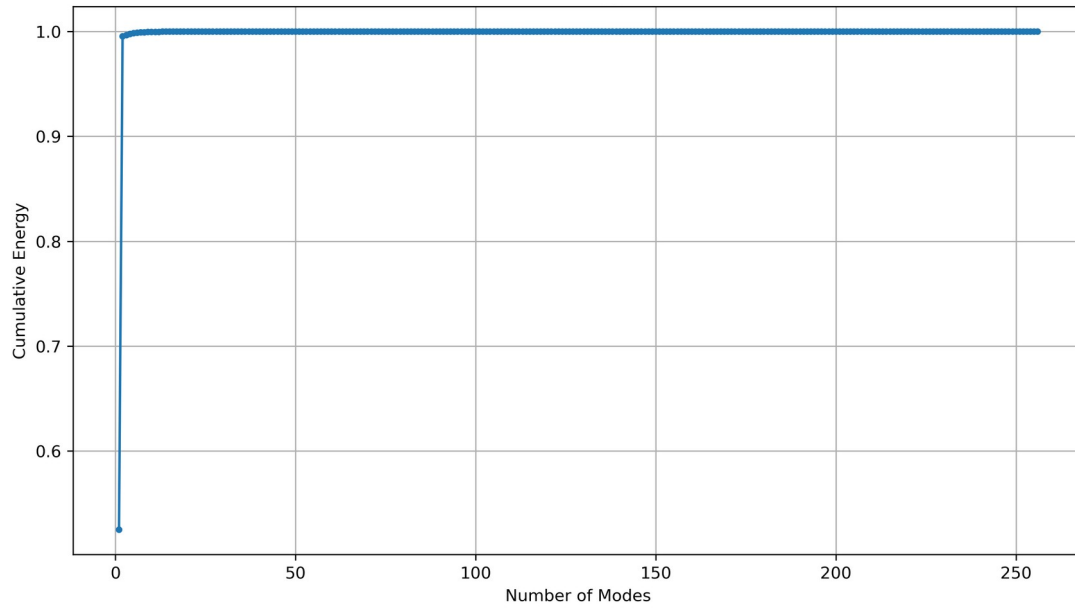
Modes	Cumulative Energy
First 2 modes	99.5%
First 7 modes	99.7 %

Table 1: Energy Contribution of Modes



Figure_18. Contribution of First 10 modes

5.4.4



Figure_19.Cumulative Energy of Modes

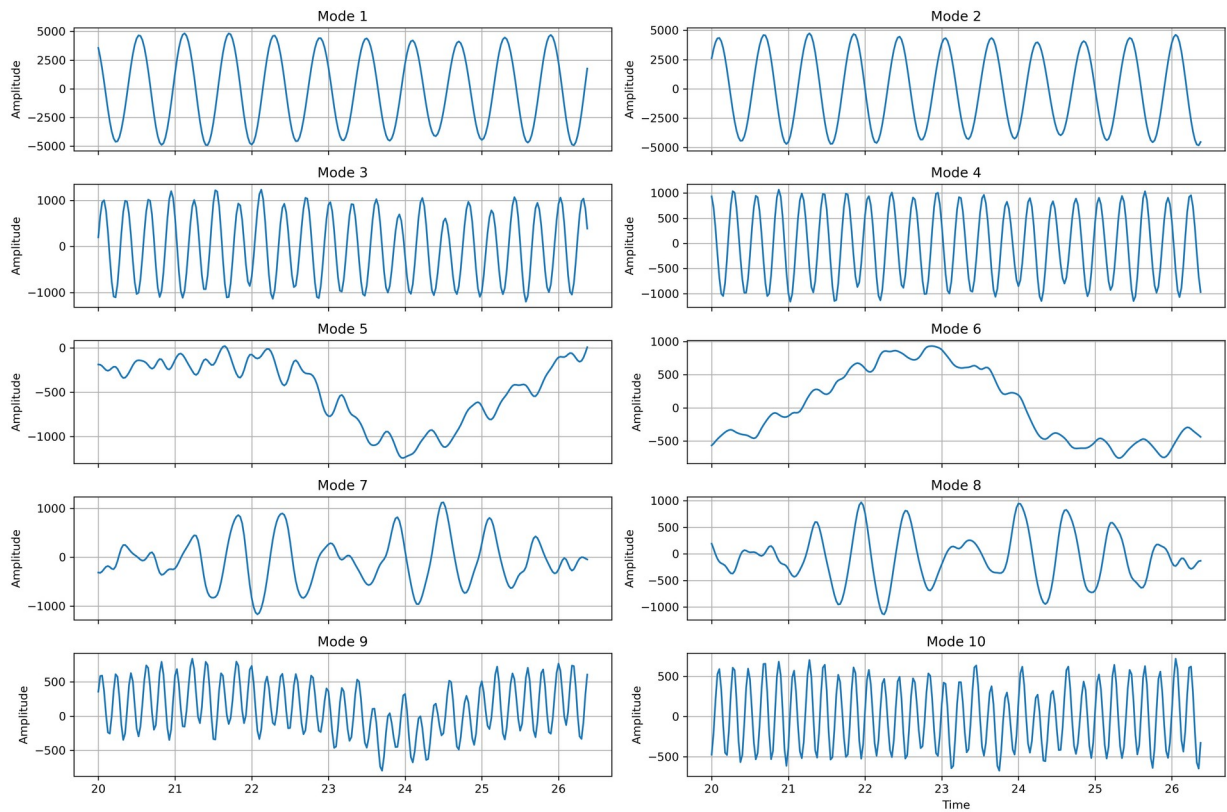


Figure 20. Amplitude of first 10 modes wrt time (for individual POD modes)

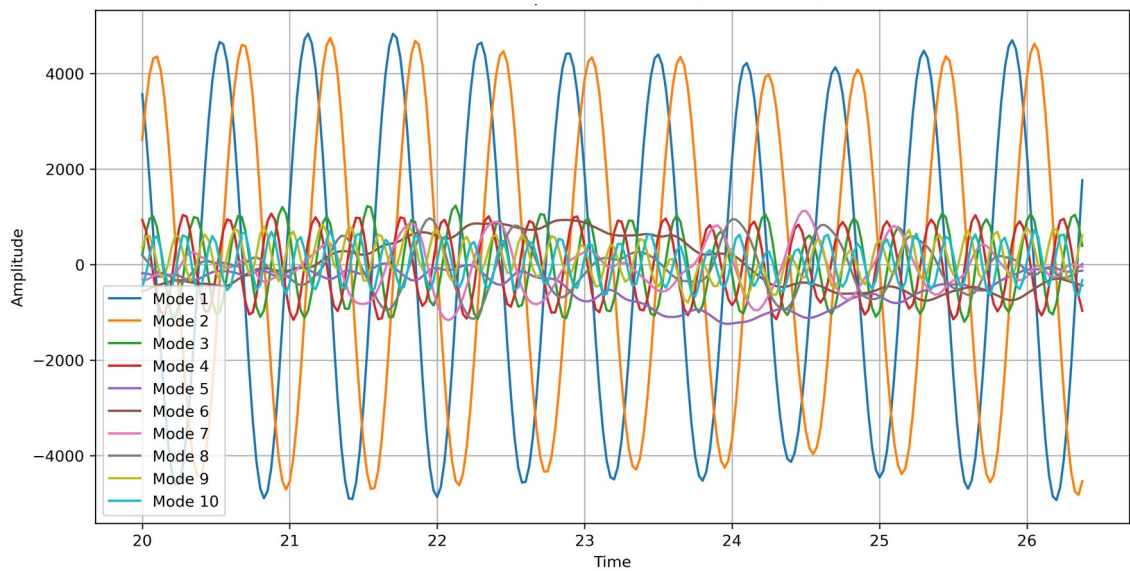


Fig 6.

Figure_21.Amplitude of first 10 modes wrt time (overlapped)

6. Conclusion:

In this study, a comprehensive numerical investigation of airflow around a bluff-body configuration—modeled after the Volvo bluff-body experiment—was carried out using OpenFOAM. The objective was to evaluate the predictive performance of two widely used Reynolds-Averaged Navier-Stokes (RANS) turbulence models, namely the **k- ϵ** and **k- ω SST** models, for the **non-reacting flow case**. Both models were assessed using detailed quantitative metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), and compared against experimental measurements.

The **k- ω SST model consistently outperformed** the k- ϵ model across all evaluated parameters including mean axial velocity, root mean square velocity, and Reynolds stress profiles. Its better near-wall treatment and blending of k- ϵ behavior in the far field contributed to more accurate flow predictions, particularly in the wake and recirculation regions.

Adaptive Mesh Refinement (AMR) based on vorticity magnitude was successfully employed to enhance resolution in dynamically active flow zones, improving the quality of the simulation without significantly increasing computational cost.

To explore the dominant coherent structures and reduce flow complexity, **Proper Orthogonal Decomposition (POD)** was applied to the time-resolved velocity field. The first few POD modes captured a significant portion of the total flow energy, indicating a strong presence of low-dimensional behavior in the unsteady wake. Mode decomposition and flow reconstruction from limited modes demonstrated the potential of POD in building reduced-order models (ROMs) that retain essential dynamic features while being computationally efficient.

Overall, this study not only highlights the efficacy of the k- ω SST model for bluff-body flows but also demonstrates the power of combining turbulence modeling with data-driven decomposition techniques like POD. This integrated approach enhances both physical understanding and computational efficiency for complex turbulent flows.

7. Future Scope:

POD results can be validated using previous studies and experimental flow measurements (e.g., LDA/PIV data from the Volvo experiment).

- DMD can be performed to extract frequency-resolved coherent structures and compared with POD for better dynamic insight.
- Grid independence study can be conducted to ensure mesh-independent results despite the current fine resolution.
- AMR sensitivity can be analyzed by varying refinement levels and vorticity thresholds to assess their impact on flow features and POD mode accuracy.
- The setup can be extended to reacting flows to simulate lean premixed combustion and investigate flame–turbulence interactions.
- A parametric study of inlet velocity profiles and bluff-body geometry can be done to observe their effects on flow instabilities and POD modes.

8. References:

- [1] Sjunnesson, A., Henrikson, P., and Lofstrom, C., “CARS measurements and visualization of reacting flows in a bluff body stabilized flame,” AIAA paper 92-3650 , 1992.
- [2] Wu, Hao, et al. "MVP-Workshop Contribution: Modeling of Volvo bluff body flame experiment." *55th AIAA Aerospace Sciences Meeting*. 2017.